

Discrete Mathematics

Sequences and Sums

Assoc. Prof. Ferhat Uçar

Faculty of Technology Software Engineering Department

December 1, 2025

Section Overview

In this section, we will deeply explore:

- What sequences are and how to define them
- Concrete examples of sequences
- Geometric progression (multiply each time)
- Arithmetic progression (add each time)
- Strings as sequences of characters
- Recurrence relations
- Fibonacci sequence and step-by-step evaluation
- Solving recurrence relations using iteration

We will explain each concept intuitively, mathematically, and with multiple examples.

What Is a Sequence?

A **sequence** is simply an ordered list of elements.

Examples:

1, 2, 3, 5, 8

1, 3, 9, 27, 81, ...

Important points:

- The order matters.
- Repetition is allowed.
- Elements may be numbers, characters, objects, or anything.

Sequences appear in mathematics, computer science, biology, music, botany, cryptography, and countless other fields.

Formal Definition of a Sequence

A sequence is a function

$$f : D \rightarrow S$$

where:

- D is usually $\{0, 1, 2, 3, \dots\}$ or $\{1, 2, 3, \dots\}$
- S is the set of possible values

We denote:

$$a_n = f(n)$$

So the sequence is written as:

$$a_0, a_1, a_2, a_3, \dots$$

This means: “When the input is n , the output is a_n .”

Example Sequence

Suppose:

$$a_n = 2n + 1$$

Then:

$$a_0 = 1, \quad a_1 = 3, \quad a_2 = 5, \quad a_3 = 7$$

This generates the odd numbers.

Interpretation:

- The rule creates each term directly.
- No need to know previous terms.
- This is called an **explicit formula**.

Definition: Geometric Progression

A **geometric progression (GP)** is a sequence of the form:

$$a_n = a r^n$$

where:

- a = initial (first) term
- r = common ratio

Key idea: Multiply by r each step.

If $r > 1$, the sequence grows exponentially. If $0 < r < 1$, the sequence shrinks toward 0. If $r < 0$, it alternates in sign.

Examples of Geometric Progressions

Example 1 $a = 1, r = -1$

$$1, -1, 1, -1, 1, \dots$$

Alternates forever.

Example 2 $a = 2, r = 5$

$$2, 10, 50, 250, 1250, \dots$$

Each step multiplies by 5.

Example 3 $a = 6, r = \frac{1}{3}$

$$6, 2, \frac{2}{3}, \frac{2}{9}, \dots$$

This sequence gets smaller and approaches 0.

Example 4 $a = 4, r = 2$

$$4, 8, 16, 32, 64, \dots$$

Classic doubling pattern.

Definition: Arithmetic Progression

An **arithmetic progression (AP)** is a sequence of the form:

$$a_n = a + nd$$

where:

- a = initial term
- d = common difference

Key idea: Add a fixed number each time.

If $d > 0$, sequence increases linearly. If $d < 0$, sequence decreases linearly.

Examples of Arithmetic Progressions

Example 1 $a = -1$, $d = 4$

$$-1, 3, 7, 11, 15, \dots$$

Example 2 $a = 7$, $d = -3$

$$7, 4, 1, -2, -5, \dots$$

Example 3 $a = 1$, $d = 2$

$$1, 3, 5, 7, 9, \dots$$

Example 4 $a = 10$, $d = 10$

$$10, 20, 30, 40, \dots$$

Strings as Sequences of Characters

A **string** is a finite sequence of characters from a finite alphabet.

Examples:

- abcde (length 5)
- 10110101
- DNA sequence: ACGTAACG

The empty string is:

λ

This concept is fundamental in computer science: programming, data structures, automata theory, and compilers.

Definition: Recurrence Relations

A **recurrence relation** expresses a_n in terms of previous terms:

$$a_n = \text{expression involving } a_{n-1}, a_{n-2}, \dots$$

To fully define the sequence, we must provide:

- The recurrence formula
- Initial conditions (starting terms)

Many important algorithm complexities use recurrences:

$$T(n) = T(n-1) + T(n-2) + n$$

Example 1: Recurrence Evaluation

Given:

$$a_n = a_{n-1} + 3, \quad a_0 = 2$$

Compute:

$$a_1 = a_0 + 3 = 5$$

$$a_2 = a_1 + 3 = 8$$

$$a_3 = a_2 + 3 = 11$$

Interpretation: Each step increases the previous value by 3.

Example 2: Recurrence Evaluation

Given:

$$a_n = a_{n-1} - a_{n-2}, \quad a_0 = 3, \quad a_1 = 5$$

$$a_2 = 5 - 3 = 2$$

$$a_3 = 2 - 5 = -3$$

Because subtraction is involved, the sequence may increase or decrease unpredictably.

The Fibonacci Sequence

Defined by:

$$f_0 = 0, \quad f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2}$$

Interpretation: Each term is the sum of the two previous terms.

This sequence appears in:

- Nature (sunflowers, shells, pinecones)
- Computer science (divide-and-conquer)
- Art and architecture

Computing Fibonacci Terms

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = 2 + 1 = 3$$

$$f_5 = 3 + 2 = 5$$

$$f_6 = 5 + 3 = 8$$

This pattern continues indefinitely.

What Does It Mean to Solve a Recurrence?

“Solving” a recurrence means finding a formula for a_n that does **not** require computing previous terms.

Such a formula is called a:

closed formula

This allows us to compute a_n directly.

In this chapter, we illustrate the method of:

- Iteration (forward substitution)
- Backward substitution

More advanced techniques appear in later chapters.

Iterative Solution Example (Forward Method)

Given:

$$a_n = a_{n-1} + 3, \quad n \geq 2, \quad a_1 = 2$$

Compute:

$$a_2 = 2 + 3$$

$$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

Pattern:

$$a_n = 2 + 3(n - 1)$$

This gives the closed formula.

Iterative Solution Example (Backward Method)

We again use:

$$a_n = a_{n-1} + 3, \quad a_1 = 2$$

Backward substitution:

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\ &= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \end{aligned}$$

Eventually:

$$\begin{aligned} a_n &= a_2 + 3(n-2) \\ &= (a_1 + 3) + 3(n-2) \\ &= 2 + 3(n-1) \end{aligned}$$

Same closed formula as before.